Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

$$\underset{a}{CH_3} - \underset{b}{CH_2} - \underset{c}{CH_2} - \underset{d}{CH_2} - F$$

Arrange the hydrogens a, b, c, d, in decreasing order of their reactivities towards chlorination:

(A) $a > b > c > d$

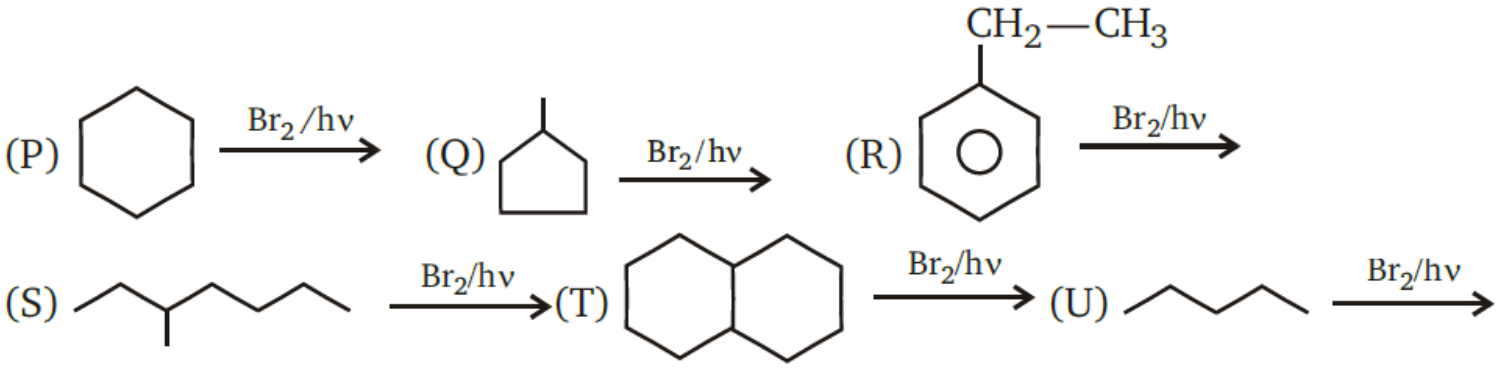
(B) $b > c > d > a$

(C) $b > c > a > d$

(D) $c > b > a > d$

Q.2

Among the following free radical bromination reactions, select those in which 2° halide is the major product -



(A) P, Q, R, S

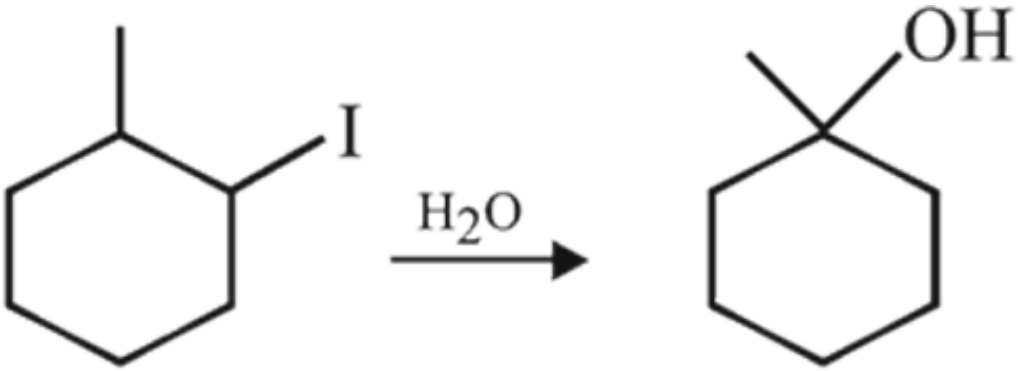
(B) P, R, U

(C) P, R, S, T

(D) P, Q, R, S, T

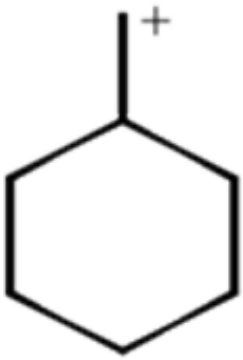
Q.3

Which of the following is not expected to be intermediate of the following reaction?

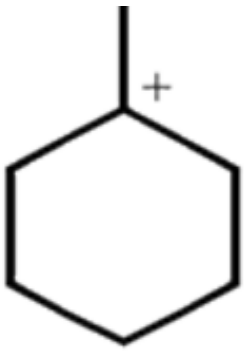


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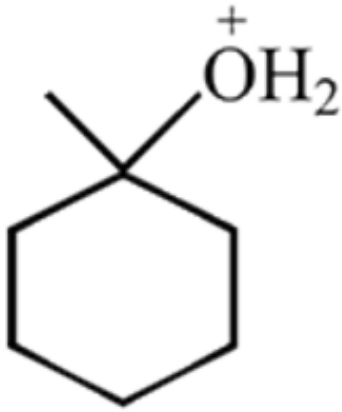
(A)



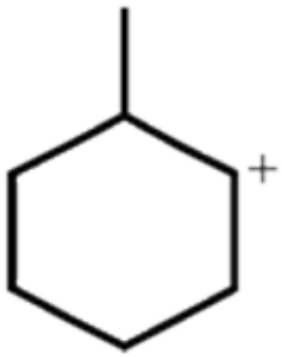
(B)



(C)



(D)



Q.4

Which of the following would not give 2-phenylbutane as the major product in a Friedel Crafts alkylation reaction on benzene ring?

(A) 1 -butene $+HF$

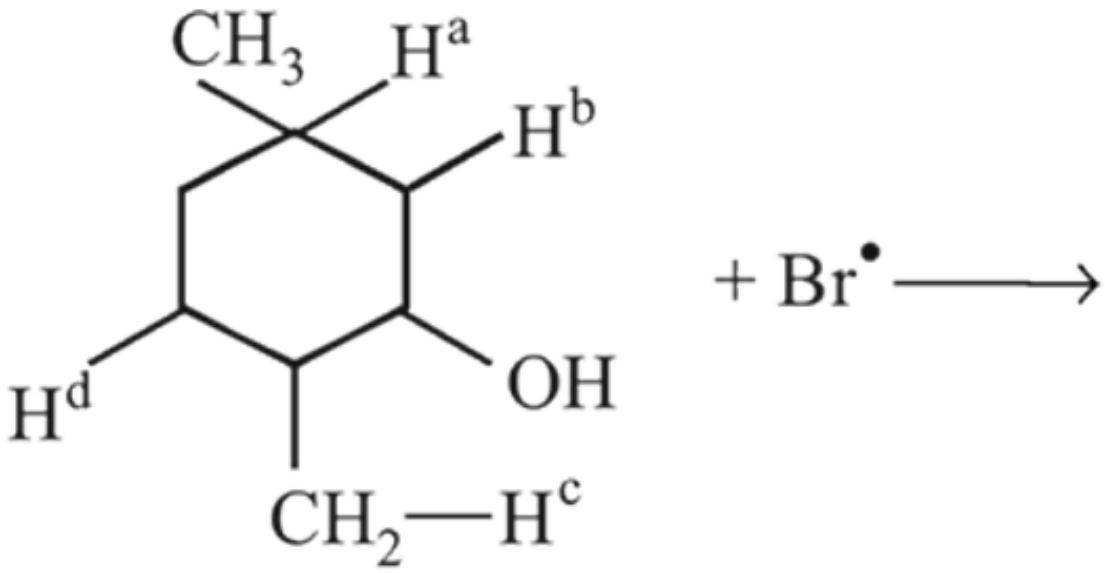
(B) 2 -butanol $+H_2SO_4$

(C) Butanoyl chloride $+AlCl_3$ then Zn, HCl

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(D) Butyl chloride $+AlCl_3$

Q.5



Br^* will abstract which of the hydrogen most readily?

(A) a

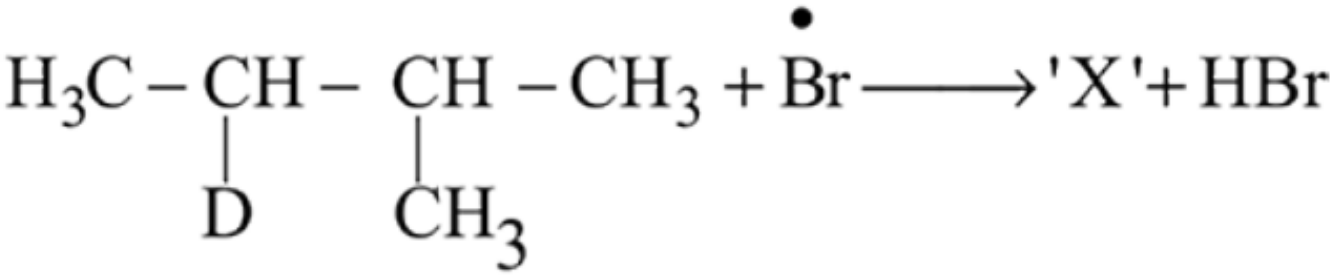
(B) b

(C) c

(D) d

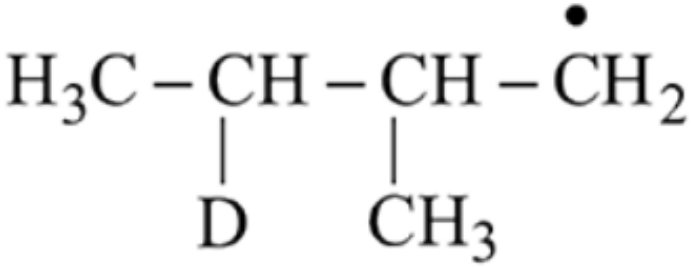
Q.6

Consider the following reaction



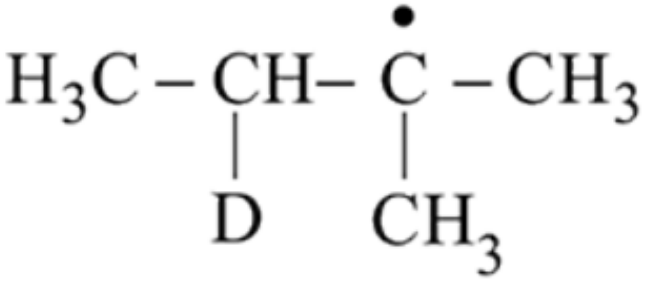
Identify the structure of the major product ' X '

(A)

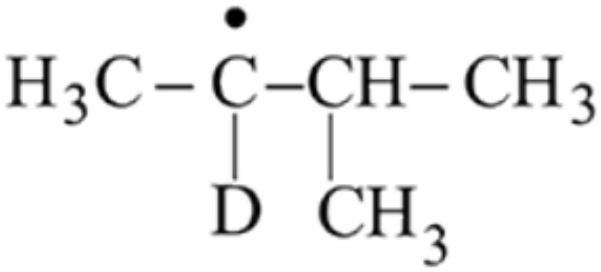


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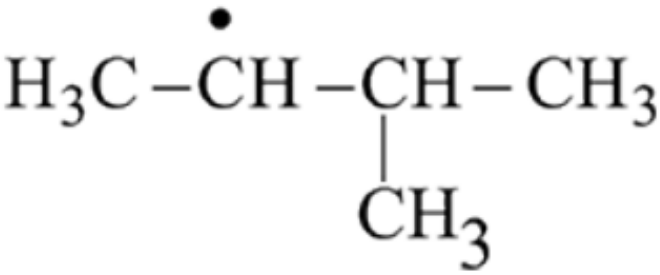
(B)



(C)



(D)



Q.7

The major product obtained in the photo catalysed bromination of 2 -methylbutane is:

(A) 1-bromo-2-methylbutane

(B) 1 -bromo-3-methylbutane

(C) 2 -bromo-3-methylbutane

(D) 2-bromo-2-methylbutane

Q.8

In preparation of alkene from alcohol using Al_2O_3 which is effective factor?

(A) Porosity of Al_2O_3

(B) Temperature

(C) Concentration

(D) Surface area of Al_2O_3

Q.9

When 1, 2 - dibromoethane reacts with sodamide giving a product X. The hybridisation state of the carbons present in it respectively, are

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(A) $sp^3,\ sp^3$

(B) $sp,\ sp$

(C) $sp^3,\ sp^2$

(D) $sp^2,\ sp^2$

Q.10

How many monochlorobutanes will be obtained on chlorination of n -butane? Consider only structural isomers.

(A) 5

(B) 2

(C) 3

(D) 4

Q.11

A hydrocarbon with formula C8H18 gives one monochloro derivative. The hydrocarbon is

(A) n-octane

(B) 2-methylheptane

(C) $2,2,4$ -trimethy pentane

(D) 2, $2,3,3$ -tetramethyl butane.

Q.12

The major product obtained in the dehydrohalogenaton of neopentylbromide with alcoholic KOHis

(A) 2 methylbut-1ene

(B) 2, 2-Dimethyl but-1-ene

(C) 2-methylbut-2-ene

(D) But-2ene

Q.13

A sample of 4.50 mg of unknown alcohol is added to CH3MgBr when 1.68 ml of CH4 at STP is obtained. The unknown alcohol is

(A) methanol

(B) Ethanol

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(C) propan-1-ol

(D) Butan-1-ol

Q.14

A hydrocarbon contains $10.5\ \text{g}$ carbon and $1\ \text{g}$ hydrogen. Its $2.8\ \text{g}$ has $1\ \text{L}$ volume at $1\ \text{atm}$ and 127°C, what is the number of carbon in the molecular formula of given hydrocarbon?

(A) 9

(B) 8

(C) 7

(D) 6

Q.15

Clemmensen reduction of a ketone is carried out in the presence of:

(A) LiAlH_4

(B) $\text{Zn} - \text{Hg}$ with HCl

(C) Glycol with KOH

(D) H_2 with Pt as catalyst

Q.16

Which one of the following reactions is expected to readily give a hydrocarbon product in good yields?

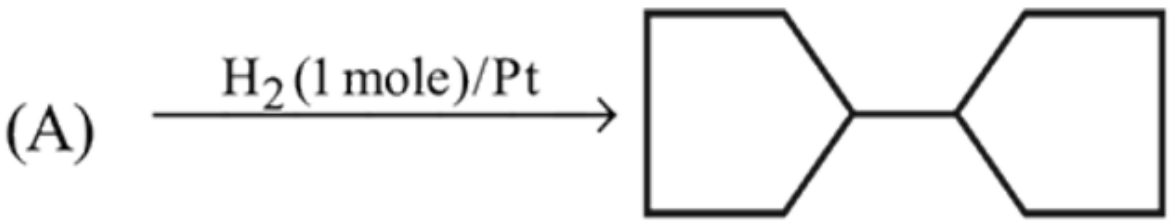
(A) $\text{RCOOK} \xrightarrow[\text{oxidation}]{\text{Electrolytic}}$

(B) $\text{RCOO}^-\text{Ag}^+ \xrightarrow{\text{Br}_2}$

(C) $\text{CH}_3\text{CH}_3 \xrightarrow[\text{hv}]{\text{Cl}_2}$

(D) $(\text{CH}_3)_3\,\text{CCl} \xrightarrow{\text{C}_2\text{H}_5\text{OH}}$

Q.17



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Double bond equivalent (degree of unsaturation) of (A) is:

(A) 1

(B) 2

(C) 3

(D) 4

Q.18

The gas liberated by the electrolysis of dipotassium succinate solution is:

(A) Ethane

(B) Ethyne

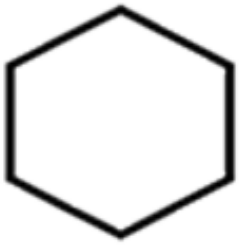
(C) Ethene

(D) Propene

Q.19

The cycloalkane having the lowest heat of combustion per CH_2 group

(A)



(B)



(C)



(D)



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Q.20

Which of the following cannot be considered as a step of mechanism in the chain reaction of methane with Cl_2?

(A) $Cl_2 \rightarrow 2\,\dot{Cl}$

(B) $CH_4 + Cl^{\bullet} \rightarrow CH_3 - Cl + H^{\bullet}$

(C) $Cl^{\bullet} + CH_4 \rightarrow \dot{C}H_3 + HCl$

(D) $Cl^{\bullet} + \dot{C}H_3 \rightarrow CH_3 - Cl$

Answers & Solutions

Q.1 Answer:

$b > c > a > d$

Solution:

Among all of them hydrogen 'b' will be most leachive towards chlorination as it is 2° hydrogen and it will get less electronegativity of fluorine atom and 'd' will be least stable as it is adjacent to the fluorine atom and does not have $3^{\circ} - 2^{\circ}$ hydrogen.

Q.2 Answer:

P, R, U

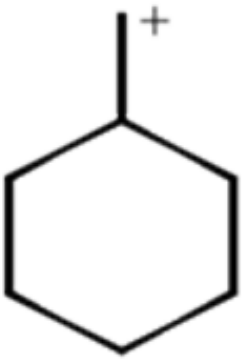
Solution:

Bromine is more selective

$\therefore$ it will form 3° halide if there is presence of 3° hydrogen.

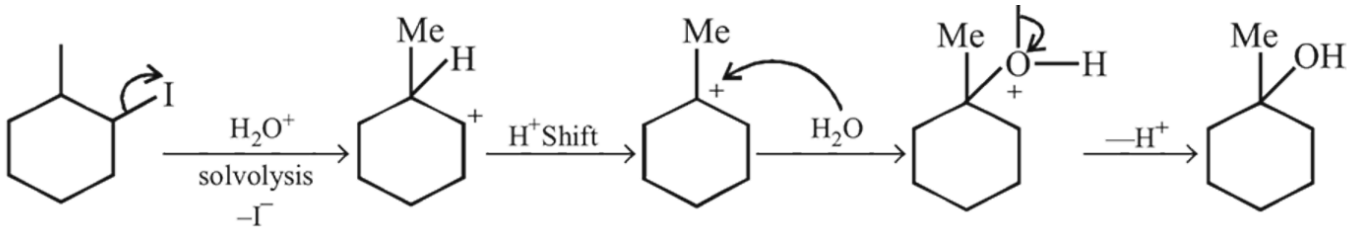
$\therefore$ Q, S, T form 3° halide as major and P, R, U form 2° halide as major.

Q.3 Answer:



Solution:

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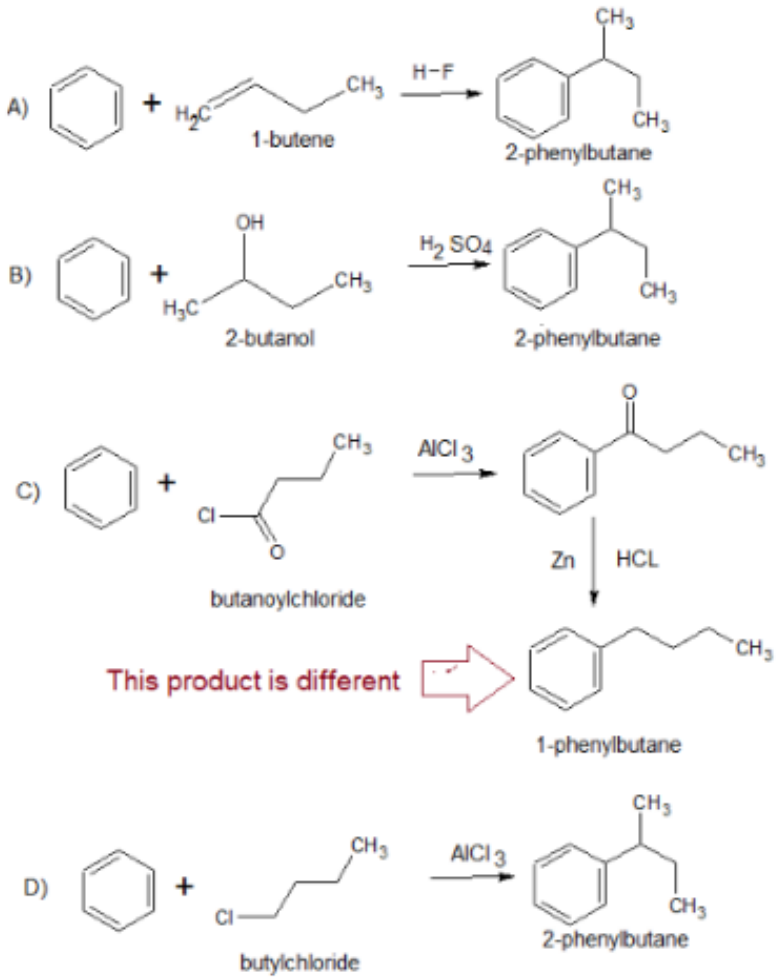


Q.4 Answer:

Butanoyl chloride $+AlCl_3$ then Zn, HCl

Solution:

In case of A, B and D stable secondary carbo cation is formed as electrophile and the products are same i.e 2-phenylbutane where as it is not possible to occur in case of (C) where the product is 1-phenylbutane.



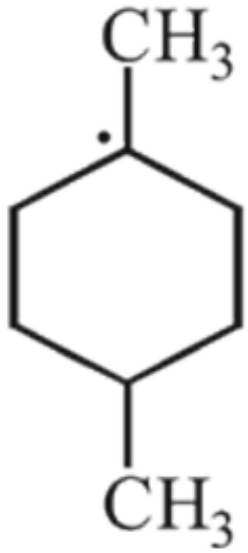
Q.5 Answer:

a

Solution:

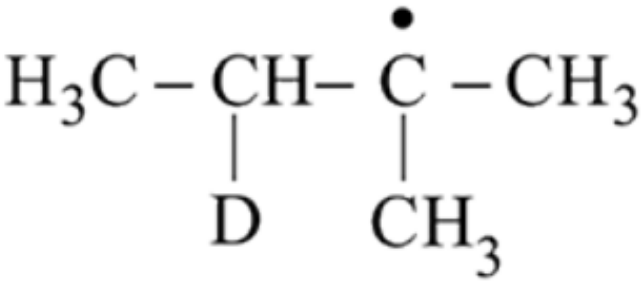
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Bromine is more selective ∴ abstract that hydrogen which forms stable free- radical



(3° free-radical) is most stable.

Q.6 Answer:



Solution:

Br° **is less reactive and more selective and so the most stable free radical** (3°) **will be the major product.**

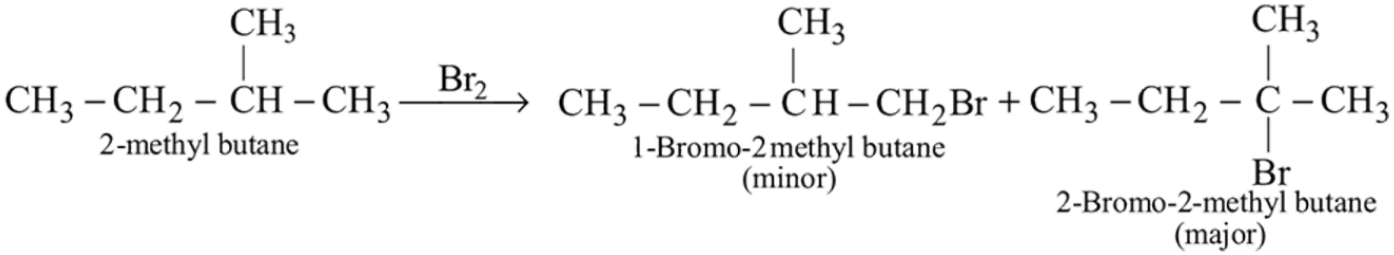
Q.7 Answer:

2-bromo-2-methylbutane

Solution:

The order of substitution in different alkanes is $3^{\circ} > 2^{\circ} > 1^{\circ}$

Thus the bromination of 2 -methyl butane mainly gives 2 -Bromo -2 **- methyl butane**



Q.8 Answer:

Temperature

Solution:

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Temperature is the effective factor for dehydration of alcohols by Al_2O_3.

$$R-CH_2-CH_2-OH \xrightarrow[350^\circ-380^\circ C]{Al_2O_3} R-CH=CH_2+H_2O$$

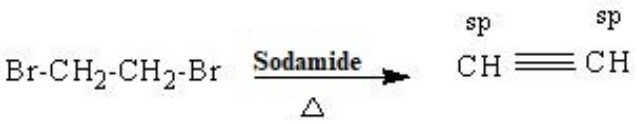
At $220^\circ-250^\circ$ it forms ether.

Q.9 Answer:

sp, sp

Solution:

Alkyl halides react with sodamide in an elimination to yield an alkene or alkyne (from dihalides). For instance,



1, 2 - **dibromo ethane acetylene**

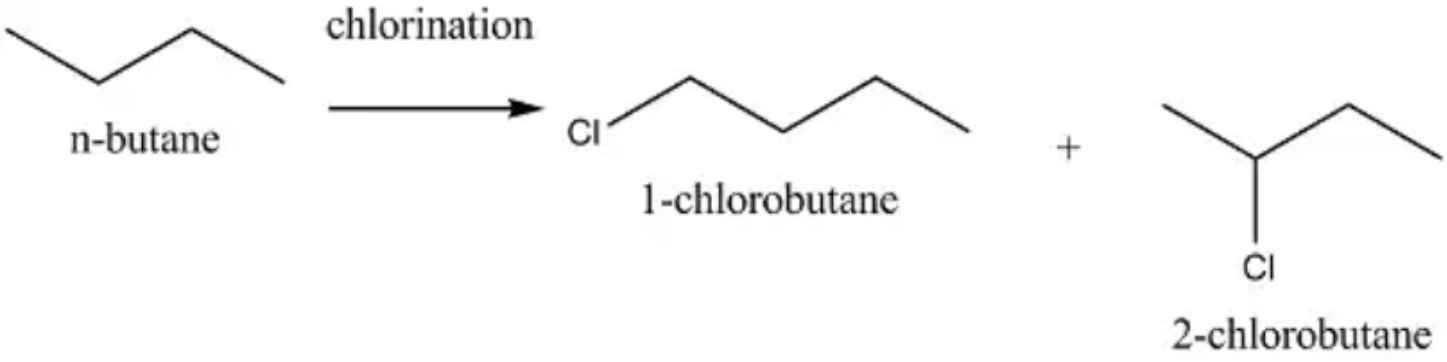
Hence, product has both sp**-hybridised carbon.**

Q.10 Answer:

2

Solution:

Two monochlorobutanes (structural isomers) will be obtained on chlorination of n-butane.



Q.11 Answer:

2, 2, 3, 3 -tetramethyl butane.

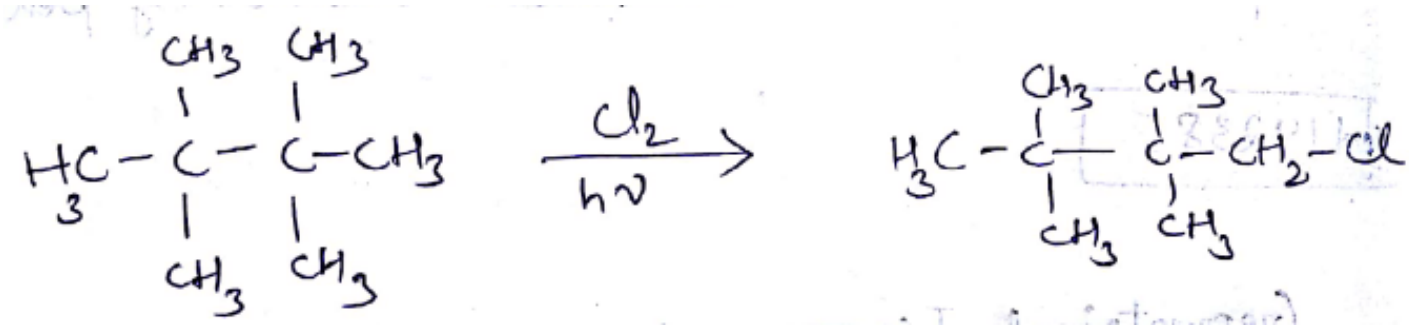
Solution:

$C_8H_{18} \xrightarrow{Cl_2}$ **one monochlorintaion product.**

From, the given reaction we can say that this 2, 2, 3, 3 **tetramethyl butane give only one chlorination**

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product as chlorine is added to one of the $-CH_3$ group.

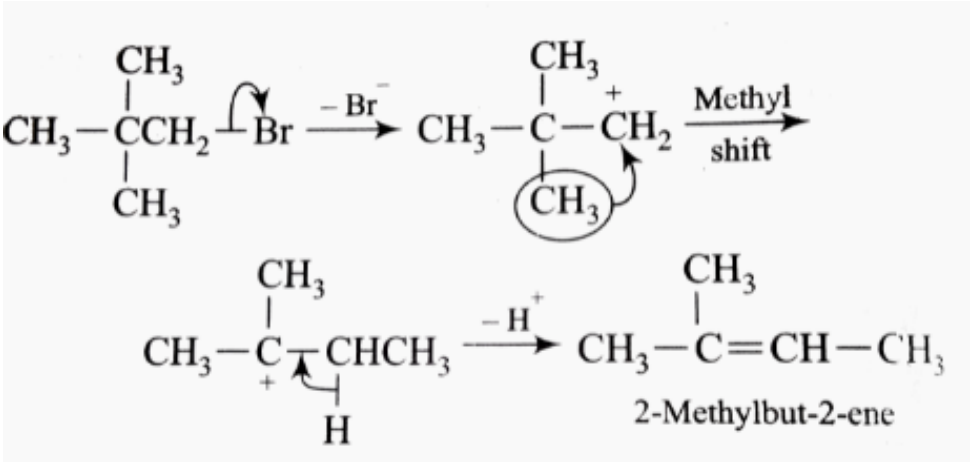


Q.12 Answer:

2-methylbut-2-ene

Solution:

In this reaction, 1° carbonium ion is formed which rearranges to form 3° carbonium ion. Hence, 2-methyl-2-butene is formed as the main product.



Q.13 Answer:

propan-1-ol

Solution:

$$R-OH + CH_3MgBr \longrightarrow CH_4 + \begin{matrix} OR \\ | \\ Mg \\ | \\ Br \end{matrix}$$

If M is the molecular weight of alcohol

$$\frac{4.50}{M} = \frac{1.68}{22.4}$$

M = 60, Therefore, the unknown alcohol is propan-1-ol.

Q.14 Answer:

7

Solution:

Sum of masses of C **and** $H = 10.5 + 1.0 = 11.5$

% $\text{of } C = \frac{10.5}{11.5} \times 100 = 91.3\%$

% $\text{of } H = \frac{1.0}{11.5} \times 100 = 8.7\%$

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Element $\%$ **Ratio of atoms** **Simplest ratio**

$\mathrm{C} \quad 91.3 \quad 91.3/12 = 7.61 \quad 7.61/7.61 = 1 \times 7 = 7$

H

$8.7 \quad 8.7/1 = 8.7 \quad 8.7/7.61 = 1.14 \times 7 = 8$

$\therefore$ **Empirical formula** $= \mathrm{C_7H_8}$

We know that $n = \frac{PV}{RT}$ $\frac{\text{mass}}{\text{molar mass}} = \frac{PV}{RT}$

$\frac{2.81}{\mathrm{M}} = \frac{1\times1}{0.082\times400}; \mathrm{M} = 92.168 \approx 92$

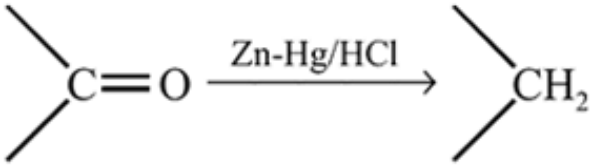
$\therefore n = \frac{84+8}{92} = 1$

$\therefore$ **Molecular formula** $= (\mathrm{C_7H_8}) \times 1 = \mathrm{C_7H_8}$

Q.15 Answer:

$\mathrm{Zn-Hg}$ with HCl

Solution:



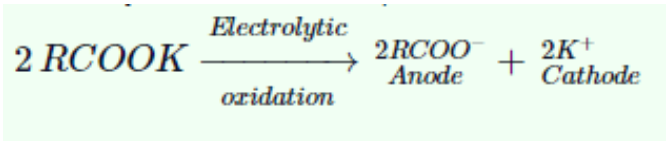
Ketone

Q.16 Answer:

$\mathrm{RCOOK} \frac{\text{Electrolytic}}{\text{oxidation}}$

Solution:

Electrolysis of sodium or potassium salt of carboxylic acid gives good yield of hydrocarbon



At anode $2RCOO^- \longrightarrow RCO\dot{O} + 2e^-$

$2RCO\dot{O} \longrightarrow R-R + 2CO_2$

At cathode $2K^+ + 2e^- \longrightarrow 2K$

$2\ \mathrm{K} + \mathrm{H_2O} \longrightarrow 2\mathrm{KOH} + \mathrm{H_2} \uparrow$

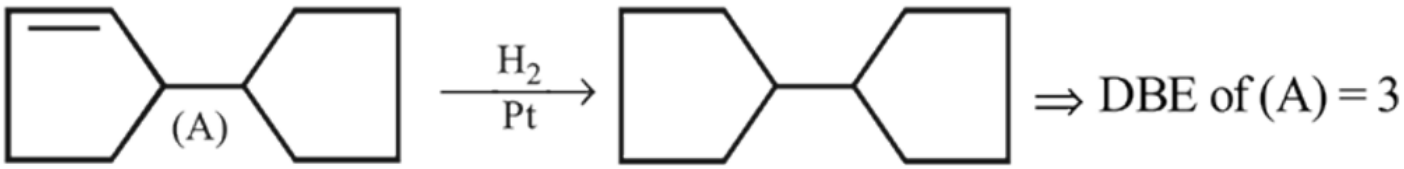
Q.17 Answer:

3

Solution:

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Molecular formula



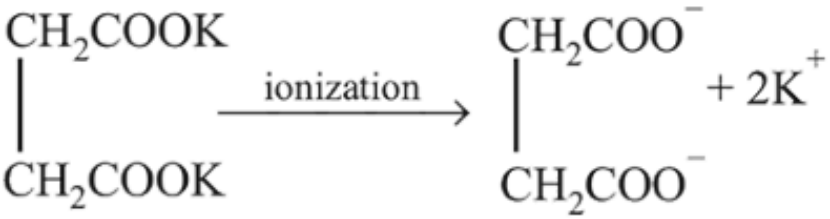
$\Rightarrow$ DBE of (A) = 3

Q.18 Answer:

Ethene

Solution:

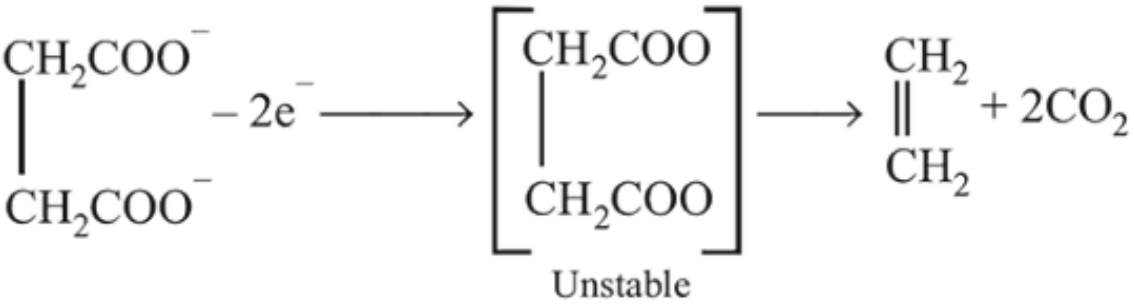
Ethene is obtained by electrolysis of dipotassium succinate as follows



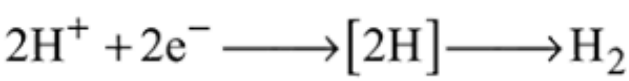
Pot. Succinate



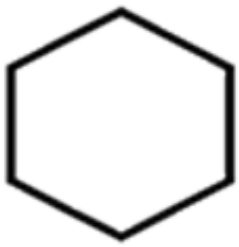
At anode :



At cathode :



Q.19 Answer:



Solution:

More is the stability, lower is the heat of combustion (See Baeyer's strain theory). Cyclohexane is more stable.

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Q.20 Answer:

$CH_4 + Cl^{\cdot} \rightarrow CH_3 - Cl + H^{\cdot}$

Solution:

Free radicals are formed as intermediates.

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